Practical No : 9

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Date:

Q.9) Write a C program to implement queue.

```
#include  <stdio.h>
#include <conio.h>
#define MAX 5
int queue[MAX];
int front = -1;
int rear = -1;
void enqueue(int value)
{
   if (rear == MAX - 1)
   {
      printf("Queue is Full\n");
   }
   else
   {
      if (front == -1)
         front = 0;
      rear++;
      queue[rear] = value;
      printf("%d inserted\n", value);
   }
}
void dequeue()
{
   if (front == -1 || front > rear)
   {
      printf("Queue is Empty\n");
```

```
    }
    else
    {
        printf("%d deleted\n", queue[front]);
        front++;
    }
}
void display()
{
    int i;

    if (front == -1 || front > rear)
    {
        printf("Queue is Empty\n");
    }
    else
    {
        printf("Queue: ");

        for (i = front; i <= rear; i++)
        {
            printf("%d ", queue[i]);
        }
        printf("\n");
    }
}
int main()
{
    enqueue(10);
    enqueue(20);
    enqueue(30);
```

```
    display();

    dequeue();

    display();

    return 0;
}
```

Output

